

CMP-04 - Intent Service

Capture attributable intent and maintain traceability from Intent through Goals, Objectives, Plans, Tasks, and outcome criteria.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 06. Authoritative state: Intent Records, Goal and Objective hierarchy, intent version lineage, acceptance and outcome criteria. Explicitly non-owned: plan execution, resource placement, judgment, action Permission decision.

Critical invariant

Execution artifacts remain attributable to an authorized Intent and cannot silently broaden its scope.

Logical interfaces

CMP-04-IF-01 / Command / Register Intent

Command `Register Intent` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-02 / Command / Revise Intent

Command `Revise Intent` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-03 / Command / Decompose Goal

Command `Decompose Goal` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-04 / Command / Retire Intent

Command `Retire Intent` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-05 / Query / Get Intent Graph

Query `Get Intent Graph` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-06 / Query / Trace Objective

Query `Trace Objective` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-07 / Query / List Outcome Criteria

Query `List Outcome Criteria` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-08 / Event / Intent Registered

Event `Intent Registered` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-09 / Event / Intent Revised

Event `Intent Revised` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-04-IF-10 / Event / Objective Retired

Event `Objective Retired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Clarifying > Accepted > Active > Satisfied, Abandoned, or Superseded. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: ambiguous principal, goal drift, orphan objective, conflicting intent version, untraceable task. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-04-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-04-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-04-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-04-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-04-TST-005 - Critical invariant. Execution artifacts remain attributable to an authorized Intent and cannot silently broaden its scope. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-04-TST-006 - Failure containment. Inject ambiguous principal, goal drift, orphan objective, conflicting intent version, untraceable task. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-04-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-04-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-04-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-04-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.